

Blackreef — AxiomIQ Fleet Overview

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Coverage: 2026-01-01 -> 2026-01-30 | Samples: 14400 | Engines: 4

Run Configuration
Version: 0.1.2

Key Changes Since Last Report

- 4 removed from fleet monitoring.
- DG1 added to fleet monitoring (priority MED).
- DG2 added to fleet monitoring (priority HIGH).
- DG3 added to fleet monitoring (priority MED).
- DG4 added to fleet monitoring (priority LOW).

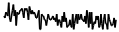
Fleet Verdict

DG2 requires inspection within 72 hours due to near-term drift. DG1, DG3 shows degradation and should be scheduled for inspection. DG4 remains healthy.

This page ranks generator sets by operational priority using health score, drift severity, and estimated time-to-limit.

Top Alerts

DG2 [HIGH] Health 64.4 | tc_lo_inlet_pressure_bar | ETA 0.1d | Inspect <72h
DG1 [MED] Health 62.7 | htcw_engine_outlet_temp_c | ETA N/A | Sched next maint
DG3 [MED] Health 63.0 | htcw_engine_outlet_temp_c | ETA N/A | Sched next maint

Engine	Trend	Health	Top Risk	Reason	Action	ETA	Pri
! DG2		64.4	tc_lo_inlet_pressure_bar	ETA 0.1d to limit (tc_lo_inlet_pressure_bar).	Inspect <72h	0.1d	HIGH
DG1		62.7	htcw_engine_outlet_temp_c	Health 62.7 (<80) and ETA unavailable; schedule inspection (htcw_engine_outlet_temp_c).	Sched next maint	N/A	MED
DG3		63.0	htcw_engine_outlet_temp_c	Health 63.0 (<80) and ETA unavailable; schedule inspection (htcw_engine_outlet_temp_c).	Sched next maint	N/A	MED
DG4		65.1	lo_inlet_temp_c	ETA 55.0d to limit (lo_inlet_temp_c).	Monitor (30d)	55.0d	LOW

Blackreef — AxiomIQ Analytics Report

Executive Summary

Engine DG2 is currently operating within defined limits, with an overall health score of 64.4 out of 100. Analysis indicates developing drift patterns rather than immediate failure conditions.

Key Developing Risk Indicators

tc_lo_inlet_pressure_bar (Turbocharger Lubrication) Trend: ↓ Observed trend suggests: Declining turbocharger lube oil pressure may suggest filter restriction or pump performance issues. Risk classification: Turbocharger bearing wear Estimated time to limits: min=0.1d max=N/A	
charge_air_pressure_bar (Air Intake / Turbocharging) Trend: ↓ Observed trend suggests: A sustained decline in charge air pressure may indicate intake restriction, air filter fouling, or reduced turbocharger efficiency. Risk classification: Performance degradation leading to thermal stress Estimated time to limits: min=25.4d max=N/A	
htcw_engine_outlet_temp_c (High Temperature Cooling Water) Trend: ↓ Observed trend suggests: Increasing HT cooling water outlet temperature may indicate reduced heat transfer, cooler fouling, or elevated engine thermal load. Risk classification: Thermal stress and reduced margin Estimated time to limits: min=237.8d max=N/A	
lo_inlet_temp_c (Lubrication Oil) Trend: ↓ Observed trend suggests: Rising lube oil inlet temperature can indicate reduced cooling efficiency, oil cooler fouling, or increased engine friction. Risk classification: Accelerated wear and oil breakdown Estimated time to limits: min=85.0d max=N/A	
engine_lo_inlet_pressure_bar (Main Lubrication System) Trend: ↑ Observed trend suggests: A gradual drop in lube oil inlet pressure can indicate filter loading, pump wear, or internal leakage. Risk classification: Loss of lubrication margin Estimated time to limits: min=N/A max=4.2d	

Operational Recommendations

- No immediate shutdown or load reduction is recommended at this time.
- Prioritize inspection of the highest-risk subsystem during the next planned maintenance opportunity.
- Monitor key lubrication and cooling parameters for continued drift over the next 72 hours.
- Re-run AxiomIQ analysis after additional data is collected (ideally 7–14 days).